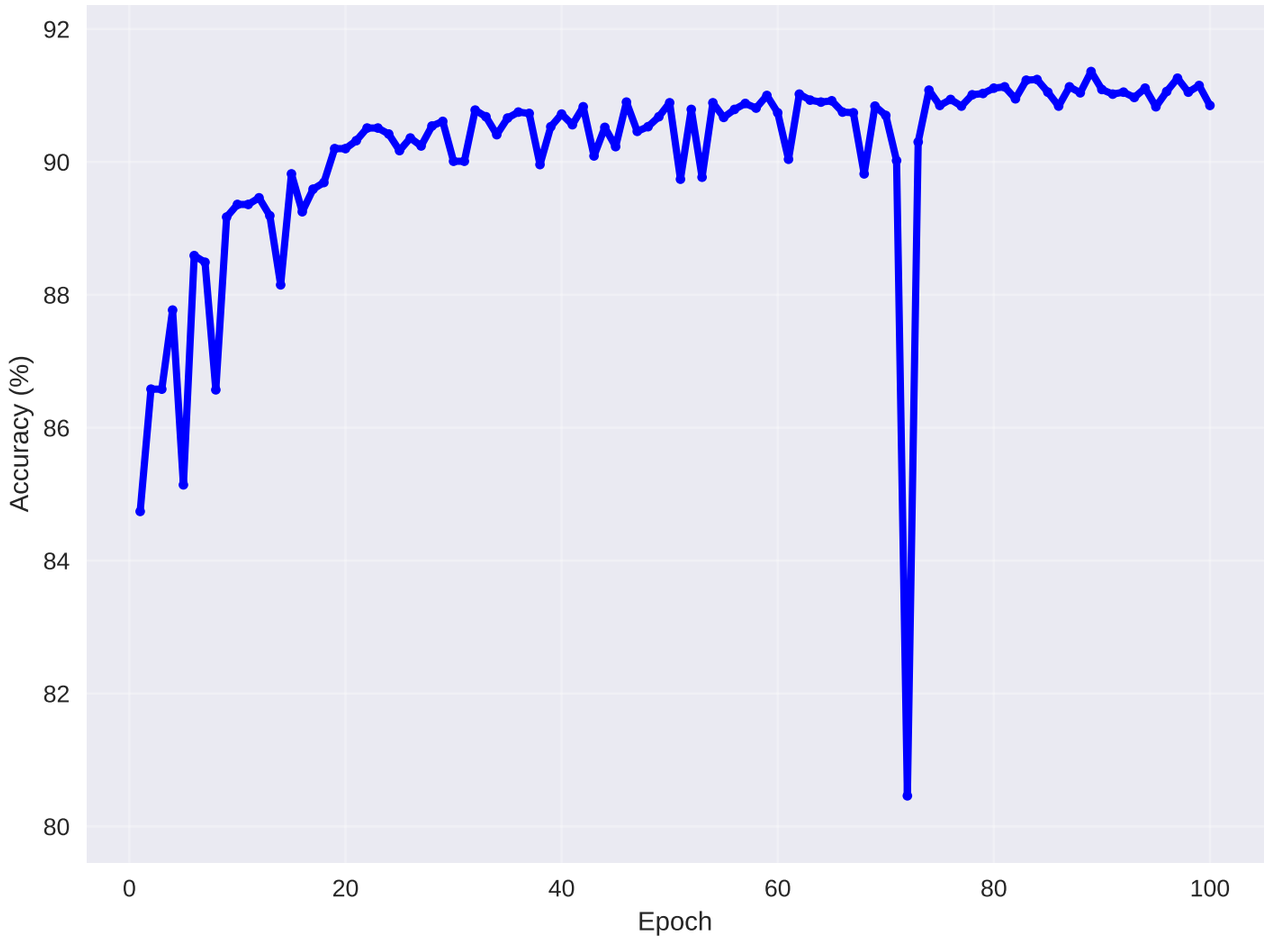
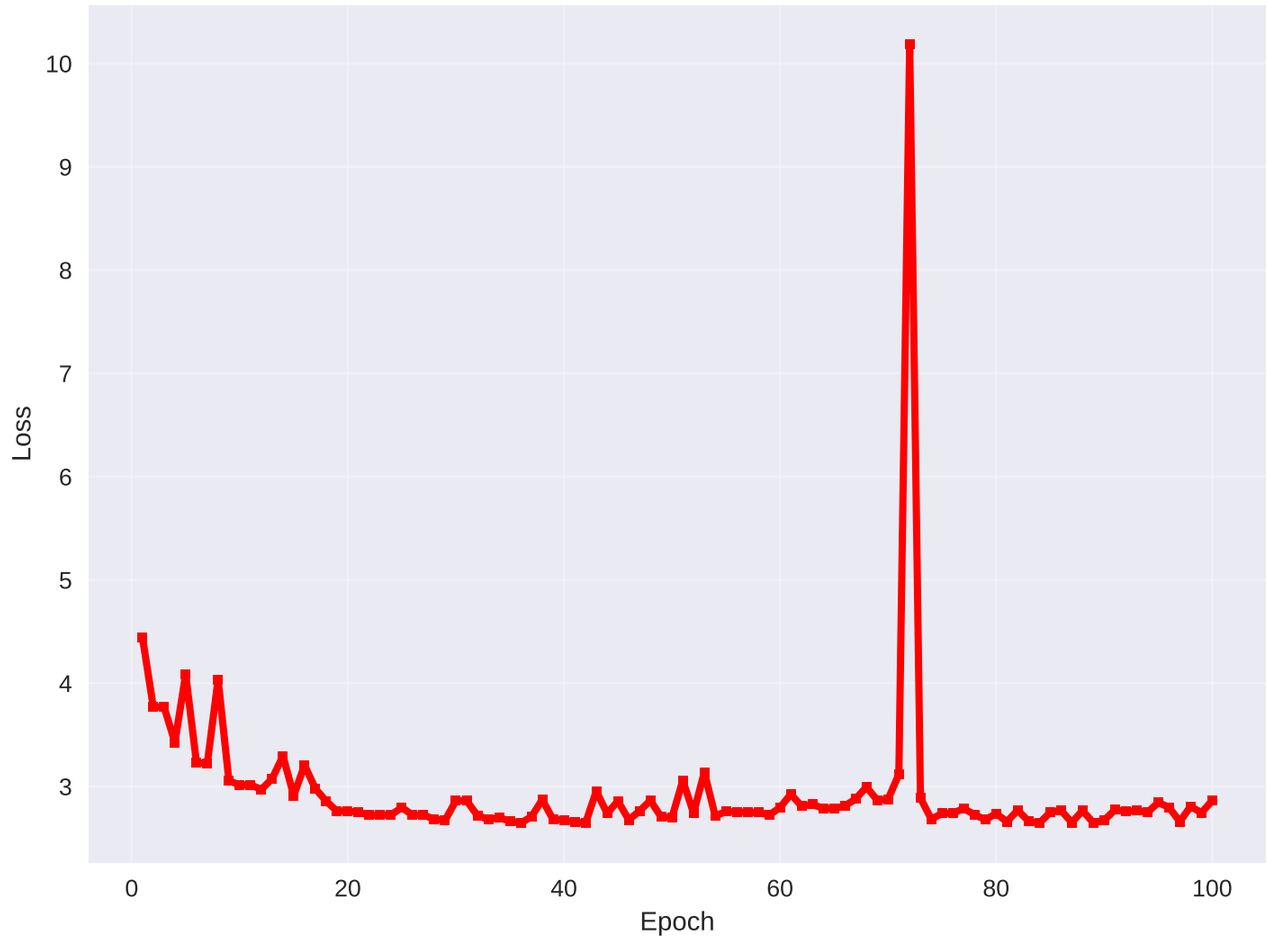
Federated Learning - Main Metrics Overview

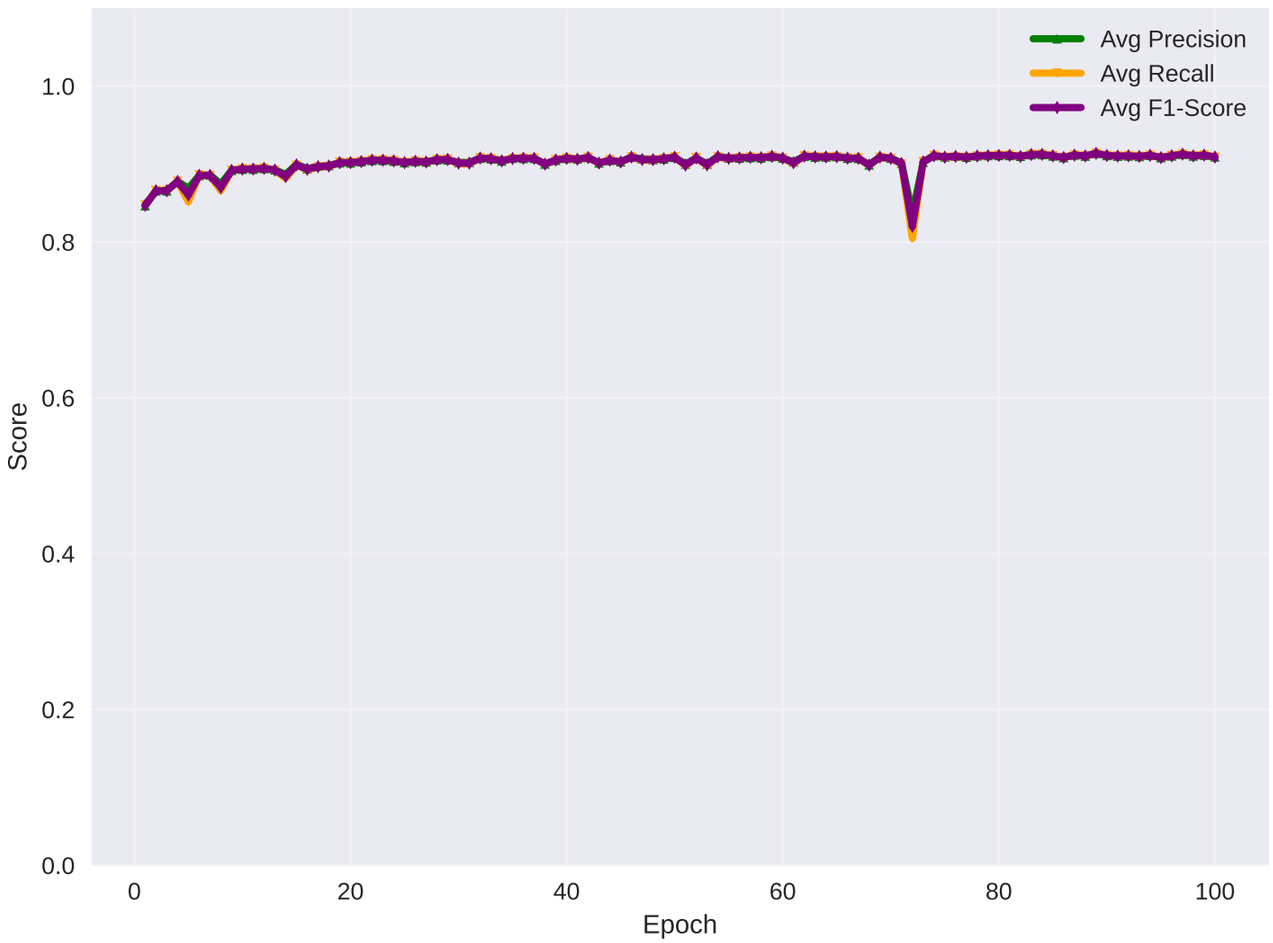
□ Global Model Accuracy Evolution



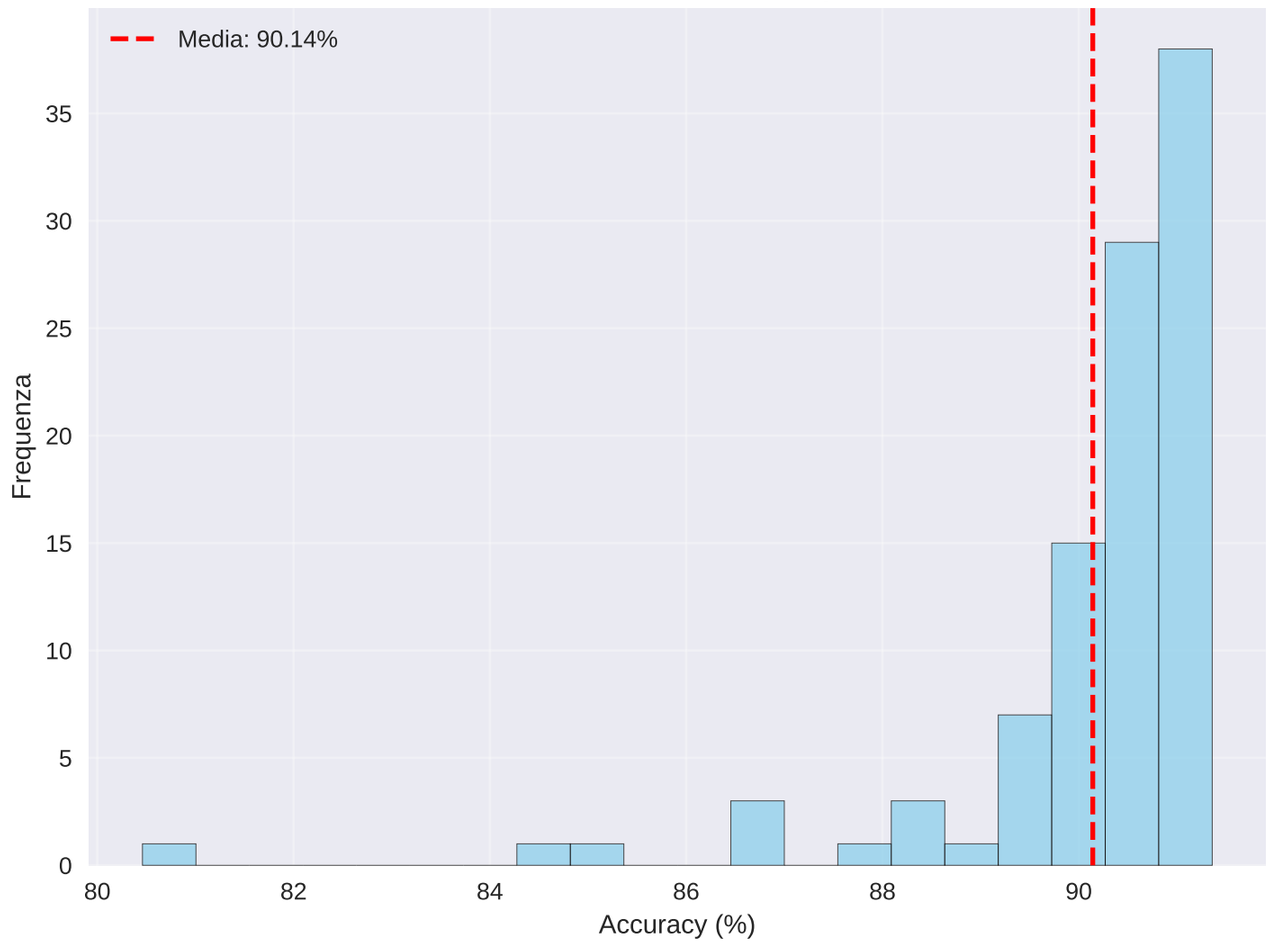
□ Global Model Loss Evolution



□ Average Precision, Recall & F1-Score

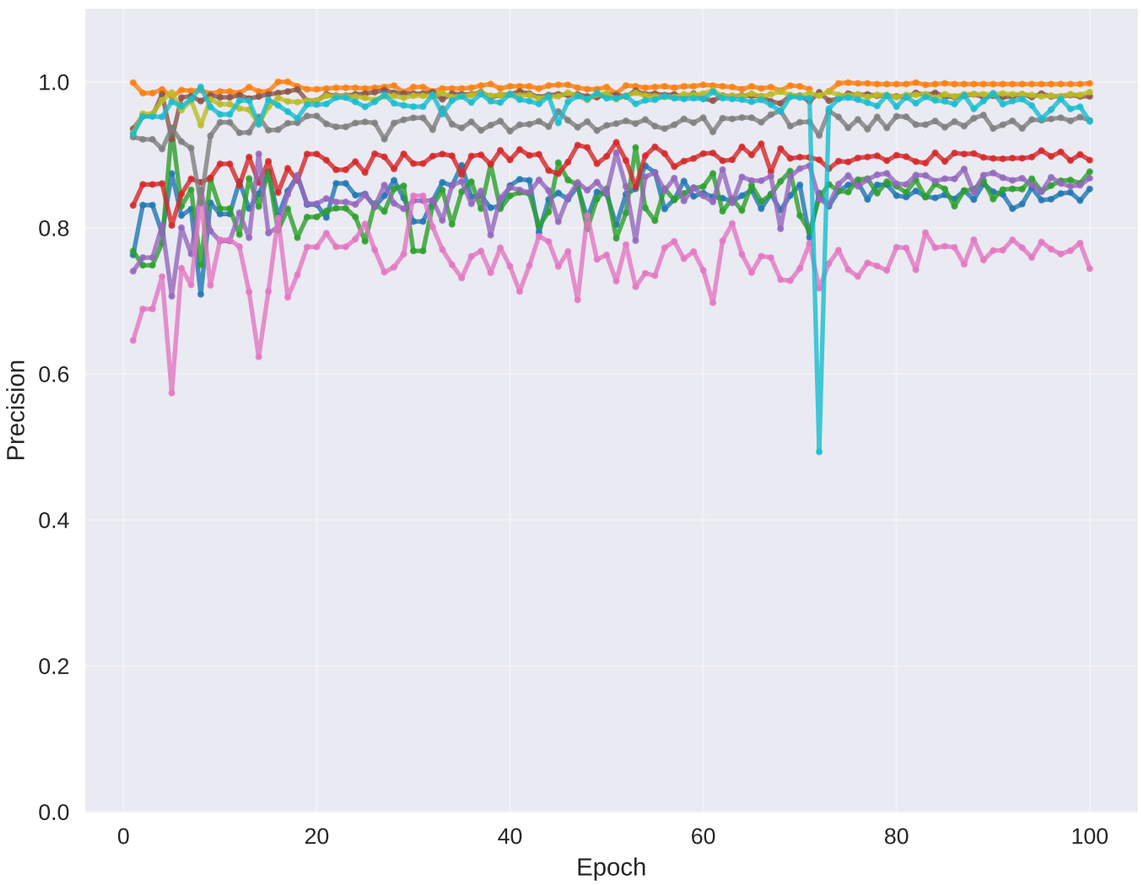


□ Distribuzione Accuracy

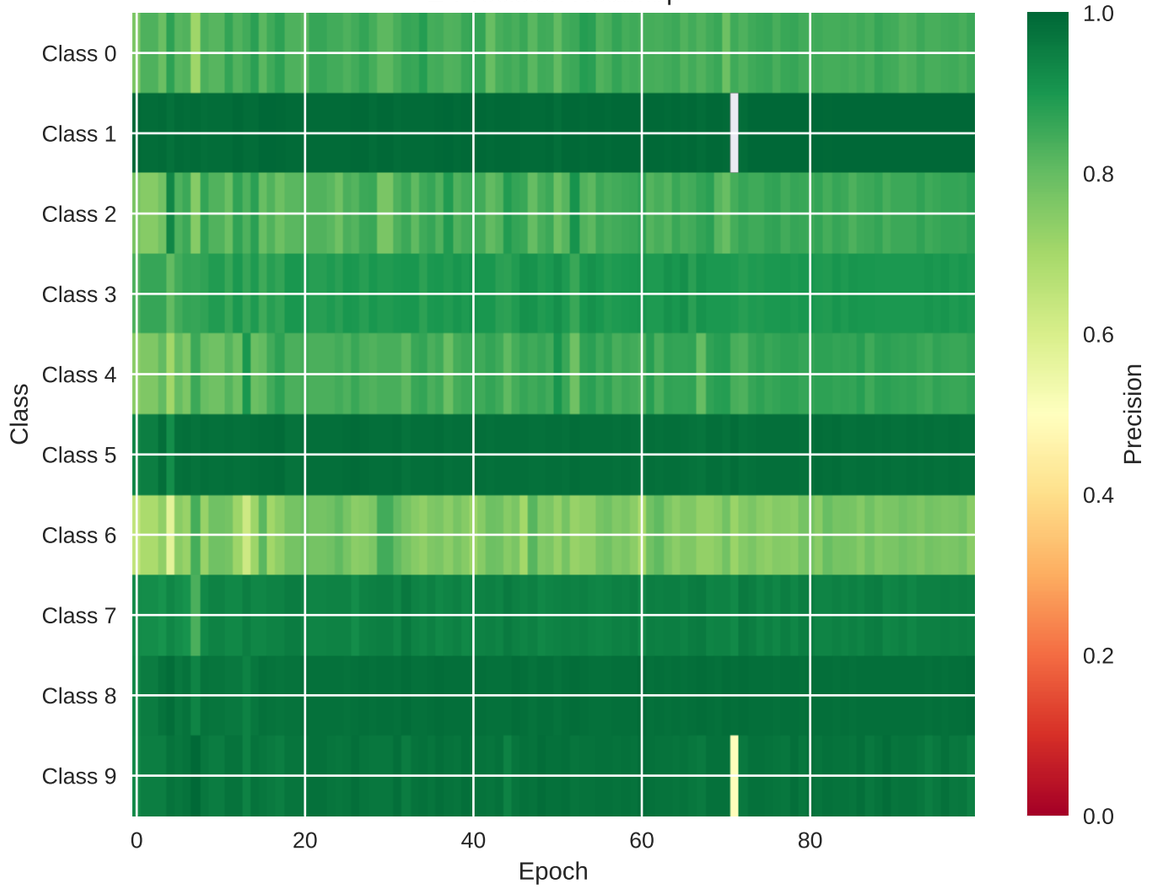


Per-Class Precision Analysis

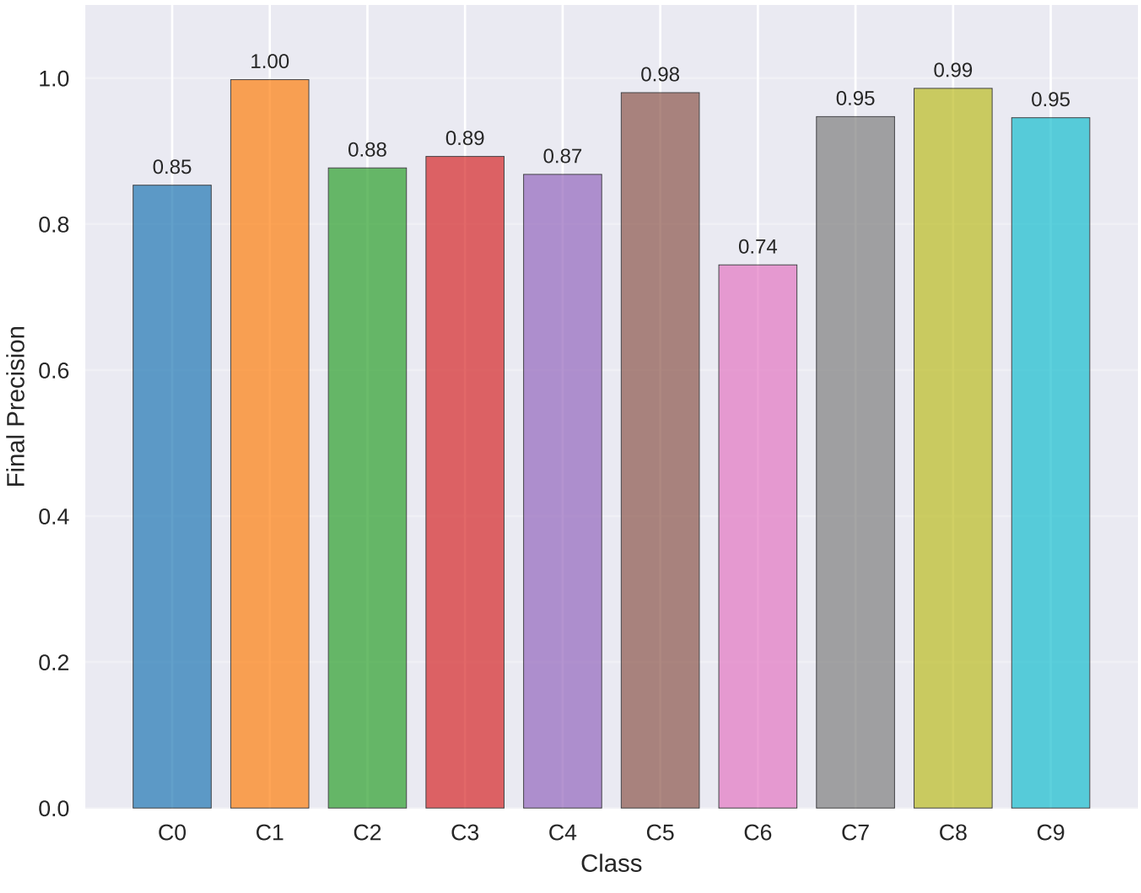
□ Per-Class Precision Evolution



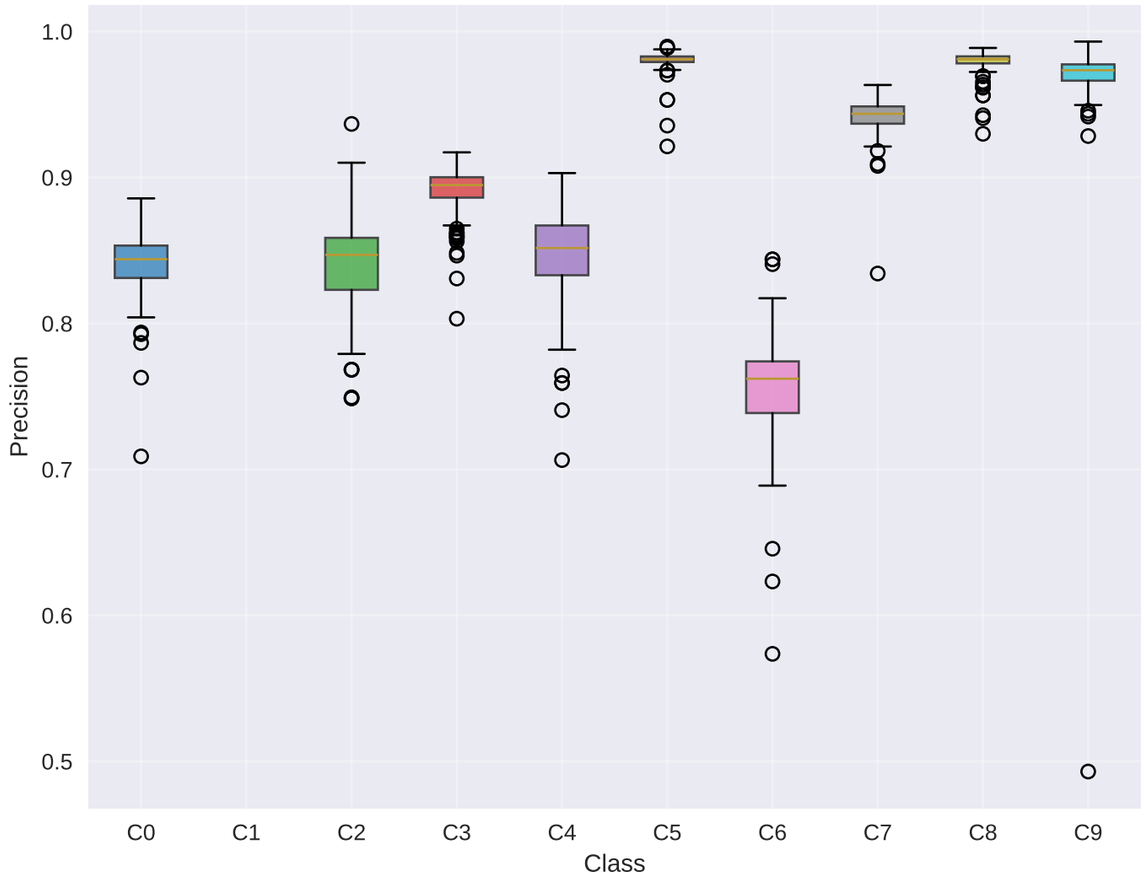
□ Precision Heatmap



□ Final Precision per Class



□ Precision Distribution per Class

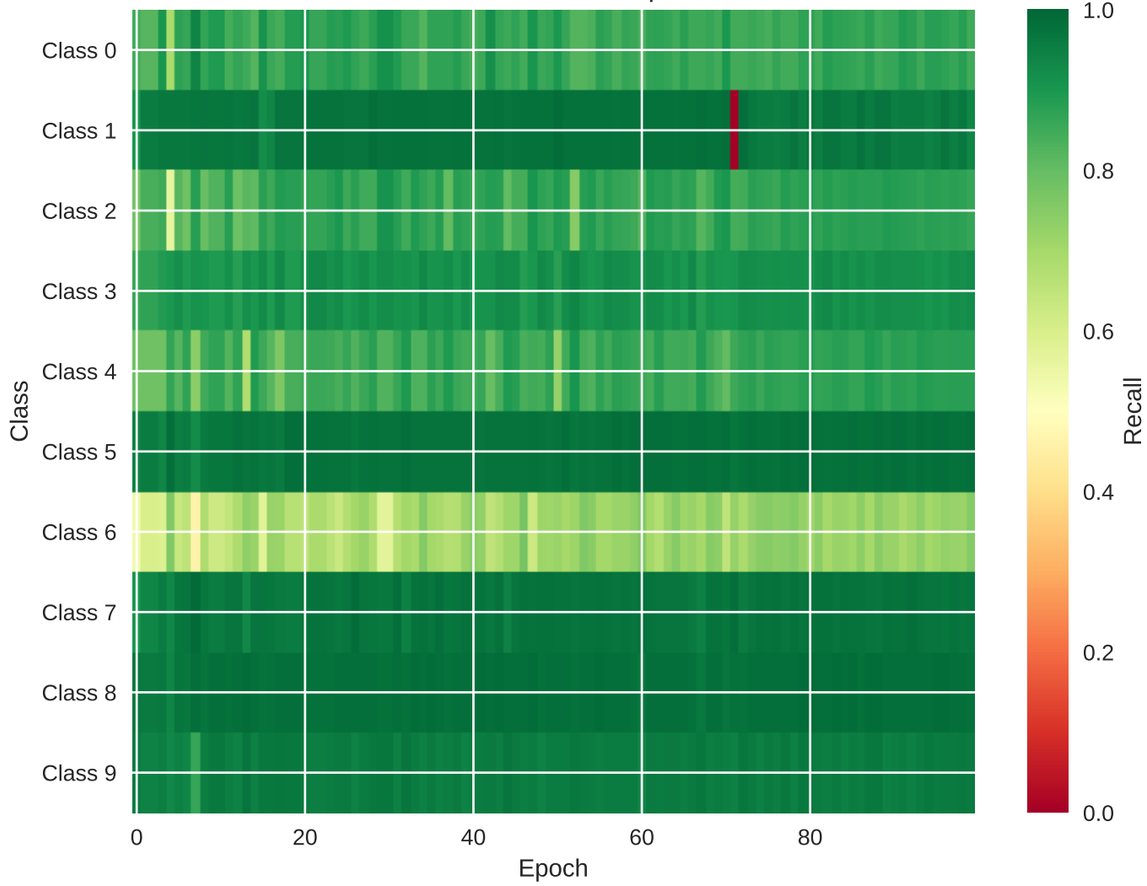


Per-Class Recall Analysis

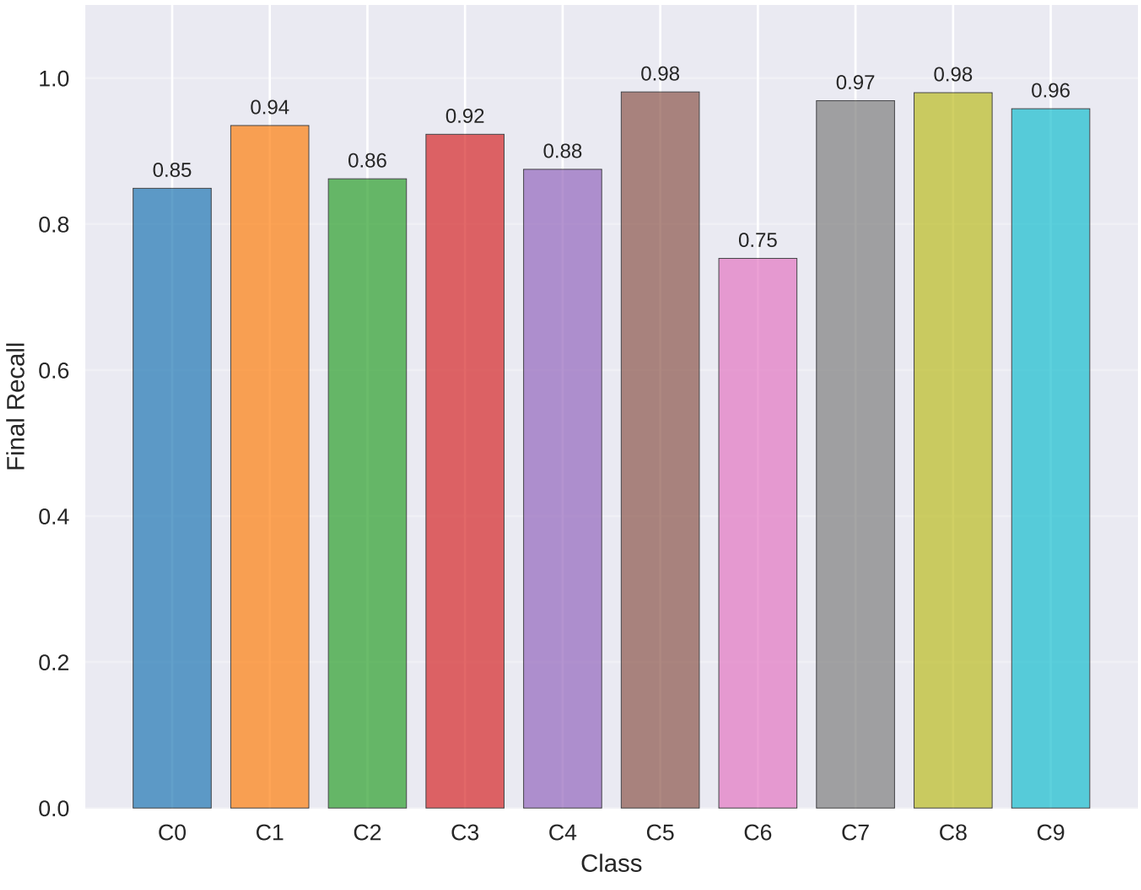
□ Per-Class Recall Evolution



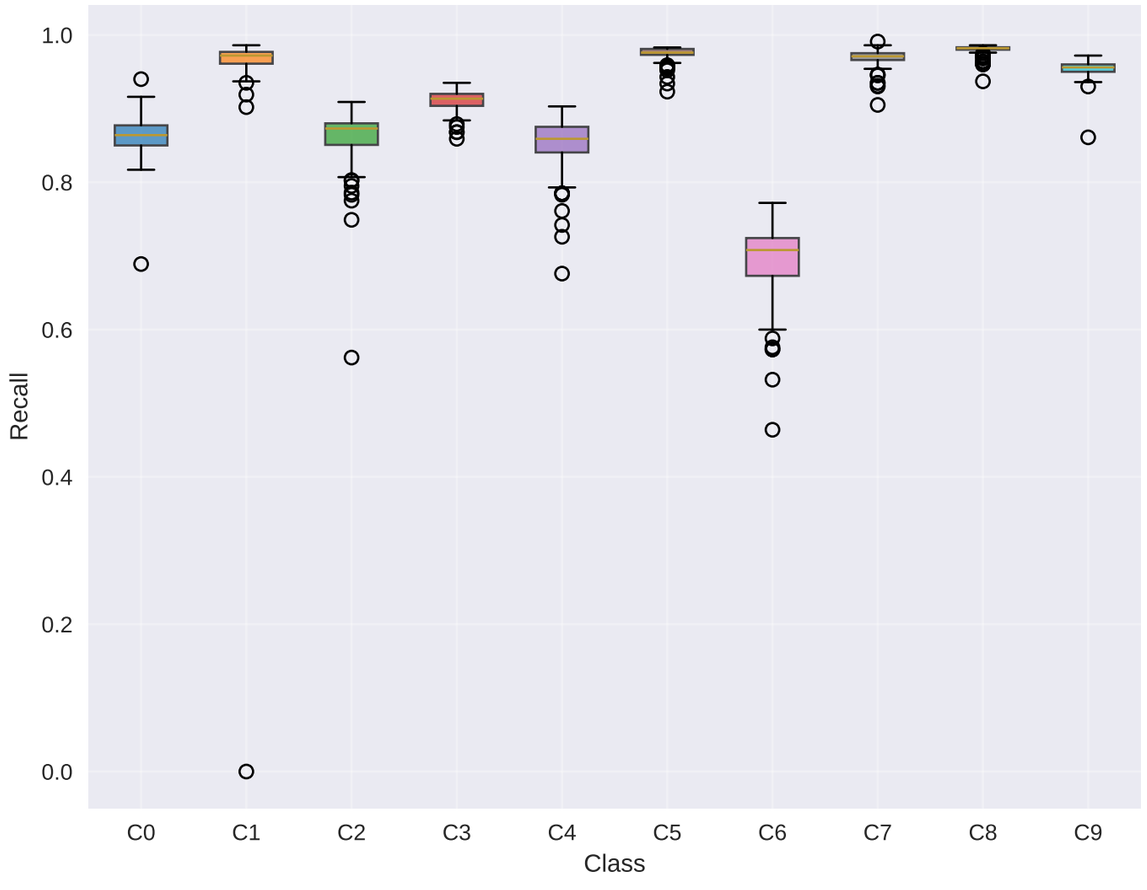
□ Recall Heatmap



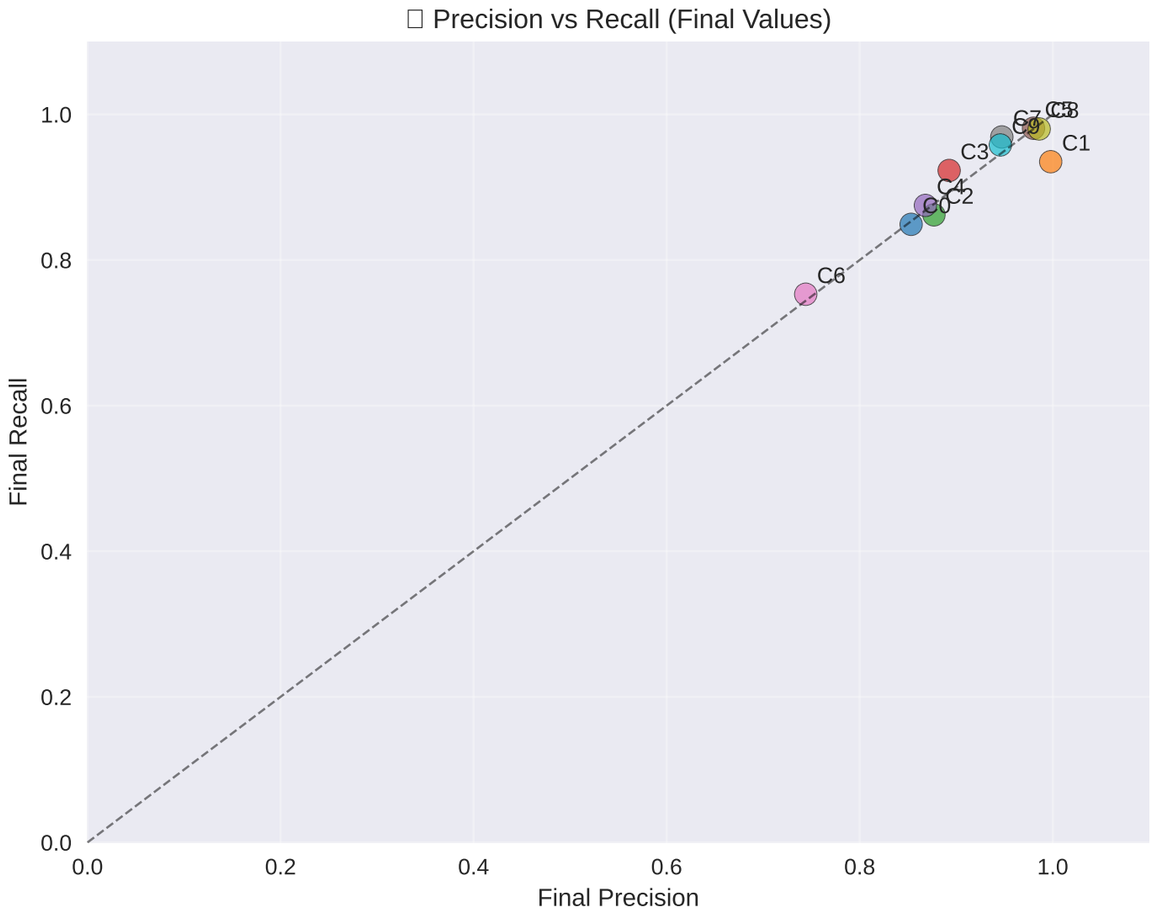
□ Final Recall per Class



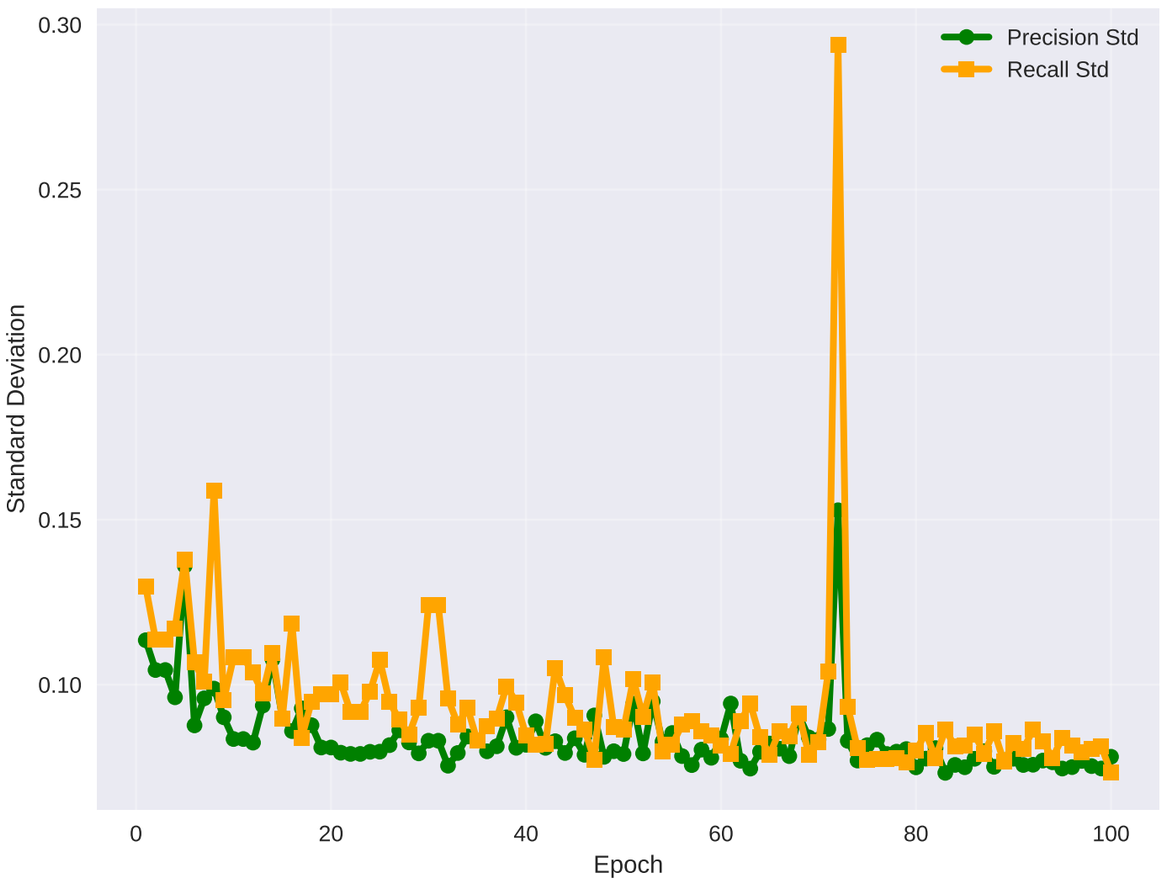
□ Recall Distribution per Class

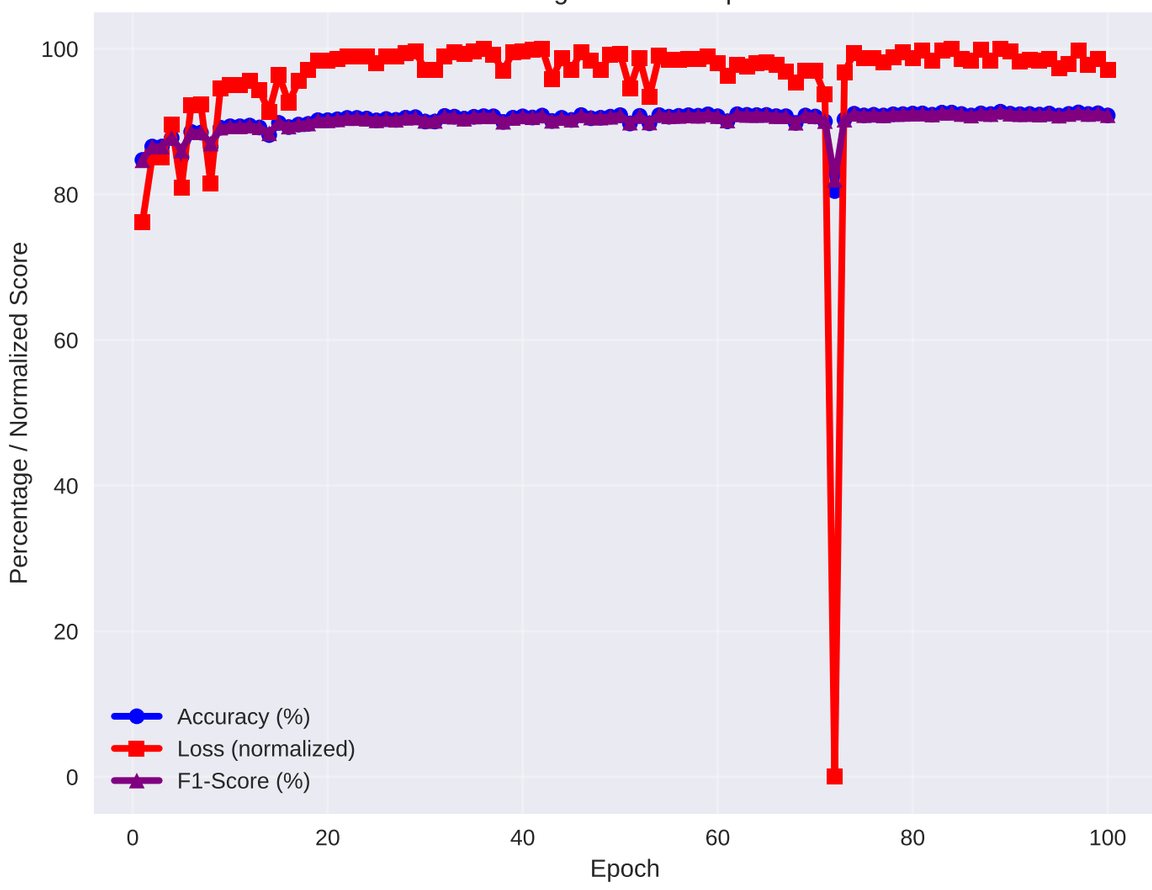


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-01 00:38:55
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 90.85%
• Loss: 2.8626
• Average Precision: 0.909
• Average Recall: 0.909
• Average F1-Score: 0.909

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.11%
• Riduzione Loss: +1.5814
• Stabilità Accuracy (CV): 0.0174

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.998)
• Peggiore Precision: Classe 6 (0.744)
• Migliore Recall: Classe 5 (0.981)
• Peggiore Recall: Classe 6 (0.753)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.853, R=0.849, F1=0.851
• Classe 1: P=0.998, R=0.935, F1=0.965
• Classe 2: P=0.877, R=0.862, F1=0.869
• Classe 3: P=0.893, R=0.923, F1=0.908
• Classe 4: P=0.868, R=0.875, F1=0.872
• Classe 5: P=0.980, R=0.981, F1=0.981
• Classe 6: P=0.744, R=0.753, F1=0.749
• Classe 7: P=0.947, R=0.969, F1=0.958
• Classe 8: P=0.986, R=0.980, F1=0.983
• Classe 9: P=0.946, R=0.958, F1=0.952

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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